

Training Leaves Traces: White-Box Model Lineage Verification using Diagonal-Dominant Fingerprints

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Abstract

Open-weight checkpoints are routinely fine-tuned, quantized, pruned, merged, and redistributed without reliable lineage metadata. Existing provenance tools are brittle: hashes fail after any weight change, metadata requires trust, and watermarks must be inserted before release. **We study a post-hoc white-box weight-lineage verification: given two checkpoints with compatible residual architecture, do they share weight-level lineage after common checkpoint transformations?** This is distinct from behavioral similarity: a distilled student imitates its teacher’s outputs, but its weights are trained from scratch and share no lineage.

We show that residual-network training induces an intrinsic fingerprint: correctly paired residual-branch products develop a diagonal-dominant fingerprint, measured by a trace/Frobenius score, that is absent at initialization and in non-residual controls. We use this signal to align blocks between checkpoints and compute a model-level lineage score from centered branch signatures. Across 12 architectures spanning language, vision, and speech (GPT-2 to LLaMA-2, ViT, Whisper, ResNets), the fingerprint recovers correct projection pairings with 91–100% accuracy, and the lineage score achieves complete separation: fine-tuned, pruned, quantized, and LoRA-merged checkpoints score high, while independently trained models and distilled students remain low. Unlike activation-based CKA and IPGuard, which track functional similarity, this provides a passive, retroactive provenance signal for white-box residual checkpoints.

1 Introduction

The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al. 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face 2024), many of which are fine-tuned (Devlin et al. 2019; Ouyang et al. 2022), quantized (Han, Mao, and Dally 2016; Dettmers et al. 2022), pruned (Han et al. 2015), or merged (Wortsman et al. 2022; Ilharco et al. 2023). This creates a *model supply chain* where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Provenance of AI-generated content is now treated as a first-class problem: Google’s SynthID watermarks text and images at generation time (Dathathri et al.

2024), yet no analogous mechanism exists for the models themselves. Model reuse is easy; reliable model provenance is not.

Existing provenance mechanisms fall short (Table 1). Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al. 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al. 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton, Vinyals, and Dean 2015). Activation-based similarity measures like CKA (Kornblith et al. 2019) and decision-boundary fingerprints like IPGuard (Cao, Jia, and Gong 2021) inherit this limitation. We need a passive, post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

The answer is yes. We find that training itself leaves a characteristic fingerprint in the weights of residual networks. In these architectures (He et al. 2016), each block adds a learned transformation to its input via a skip connection (Figure 1a). During training, the input and output projections of each block align so that their product $W_{\text{out}}W_{\text{in}}$ develops diagonal-dominant structure that is absent at random initialization and does not appear in networks without skip connections (Figure 1b). This structure emerges from the optimization process: it is learned, not an initialization artifact (Section 3).

We exploit this fingerprint for provenance verification. Each residual block yields a signature: its branch product

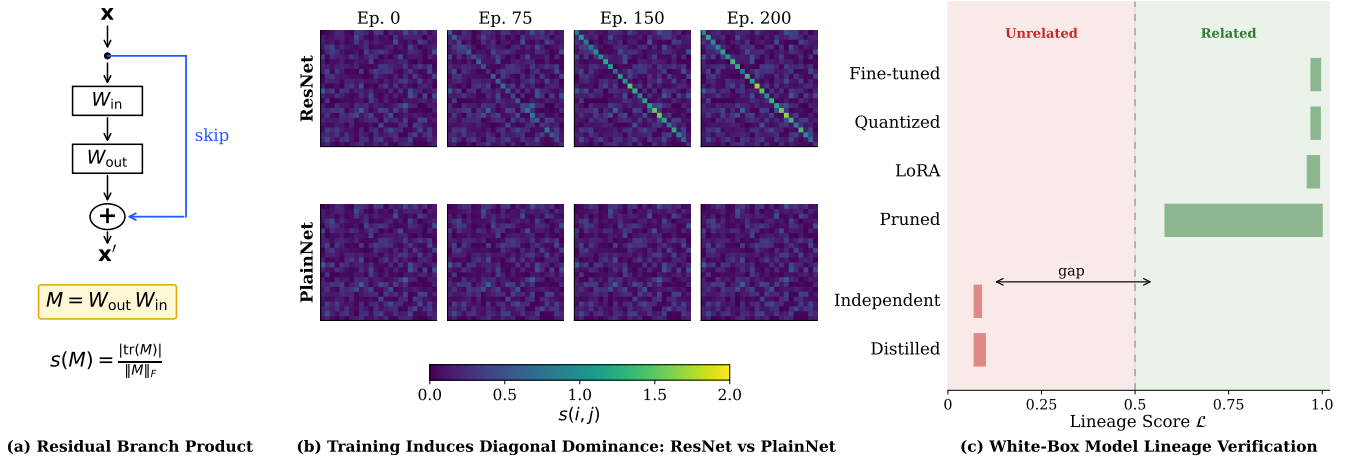


Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}}W_{\text{in}}$ for a block with skip connection; the diagonal-dominance score $s(M)$ measures trace concentration. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 5. ResNet (top row) develops strong diagonal structure (100% block-pairing accuracy); PlainNet without skip connections (bottom row) remains random (3% accuracy). The fingerprint is learned, not an initialization artifact. (c) Lineage verification: related checkpoints (fine-tuned, quantized, pruned, LoRA-merged) score high on $\mathcal{L}$; unrelated checkpoints (independent, distilled) score low, enabling clean separation above/below threshold.

scored for diagonal concentration via a trace/Frobenius ratio (Equation 3). Correctly paired blocks from the same trained model score high, while mismatched or unrelated blocks do not. The extraction is architecture-aware, respecting the specific structure of Transformer MLPs, attention paths, and ResNet bottleneck blocks (Table 2). To compare two checkpoints, we align their blocks using these signatures, aggregate the pairwise similarities into a model-level lineage score, and threshold it against a null distribution built from unrelated models (Equation 7).

The fingerprint is not specific to one model family. We observe it in public checkpoints across language, vision, and speech (GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, and ResNets) where it recovers block correspondences with over 91% accuracy (Table 3). Turning to the verification task itself, we construct controlled lineage benchmarks on MLPs and ResNet-18: models with genuine weight ancestry (produced by fine-tuning, weight noise, pruning, quantization, or LoRA merging) are cleanly separated from those without (independent training runs and distilled students), with AUROC = 1.000. The distillation case is instructive: a student can closely replicate its teacher’s outputs, yet our score correctly labels it unrelated, because its weights carry no trace of the teacher’s training (Figure 1c). Finally, on the public LLaMA-2 family, the method flags three real fine-tuned descendants (chat, Vicuna, CodeLlama) as related while rejecting four independently trained checkpoints, among them OpenLLaMA, which shares the architecture but inherits no weights.

Our contributions:

- **Diagonal-dominant fingerprint.** We establish that gradient-based training drives the input and output projections of residual blocks toward mutual alignment, concentrating their product along the diagonal, a property absent

at random initialization and never acquired by skip-free architectures.

- **White-box verification method.** We convert this signature into a practical test: per-block scores are matched across two checkpoints, aggregated into a single lineage statistic, and judged against a null distribution, requiring no watermark, no metadata, and no cooperation from the model’s distributor.
- **Broad empirical validation.** We evaluate on 280+ checkpoint pairs covering 12 architectures and 5 model families, from synthetic benchmarks with known ground truth to real-world derivatives of LLaMA-2, obtaining 91–100% block correspondence and AUROC=1.000.
- **Robustness and specificity.** The fingerprint persists through fine-tuning, quantization, pruning, and LoRA merging, yet vanishes for distilled students and independently trained models of identical architecture, a distinction beyond the reach of behavioral measures such as CKA and IPGuard.

2 Problem Setting and Related Work

This section formalizes the lineage verification problem and positions it against prior provenance mechanisms.

2.1 White-Box Model Lineage Verification

Scope. A verifier is given two checkpoints A and B and must determine whether they share weight-level lineage. We assume white-box access to both weight sets and compatible residual architectures (same layer count L and hidden dimension d); cross-architecture comparisons are out of scope. Metadata may be missing or adversarial. The verifier operates on weights alone: no training data, activations, or forward passes are available.

Method	Retro.	FT	Quant	Distill
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	~	~	✗
Decision-Boundary	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retro.: applicable retroactively. FT: survives fine-tuning. Quant: survives quantization. Distill: correctly classifies distilled copies as unrelated. ~: partial or unreliable.

Architecture	Residual branch $F(x)$	Product M
Transformer MLP / BasicBlock	$W_2\sigma(W_1x)$	W_2W_1
Attention V/O path	$W_O\text{Attn}(x)W_Vx$	W_OW_V
Attention Q/K path	bilinear $x^\top W_QW_K^\top x$	$W_OW_K^\top$
Bottleneck ResNet	$W_3\sigma(W_2\sigma(W_1x))$	$W_3W_2W_1$
SwiGLU MLP	$W_{\text{down}}[\sigma(W_{\text{gate}}x) \odot W_{\text{up}}x]$	$W_{\text{down}}W_{\text{up}}$

Table 2: Architecture-aware residual branch products. Each M composes the linear maps of one branch (dropping nonlinearities) so it maps the residual stream to itself. Factorization must match the architecture; incomplete products destroy the signal (Section 5).

Relatedness. Checkpoints are *related* if they share a common weight ancestor: either one derives from the other, or both derive from a shared parent, through a chain of transformations $\theta' = T(\theta, D)$, where T may include continued optimization (fine-tuning, RLHF), compression (pruning, quantization), or adapter merging (LoRA), and D is optional additional data. Checkpoints are *unrelated* if their weights were initialized independently, even when they share the same architecture, training data, task, or output behavior. Transformations with multiple parents, such as linear merges of two checkpoints, induce partial lineage between the suspect and each parent; we quantify this regime in Appendix D.5.

The verifier solves lineage verification: do A and B share weight ancestry? It returns one of three verdicts: **RELATED**, **UNRELATED**, or **INCONCLUSIVE** (insufficient evidence either way). The lineage score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$: the protocol establishes whether two checkpoints share ancestry, not which is the ancestor; directional attribution requires metadata or additional candidates (Appendix C). As a by-product, the verification pipeline recovers block-level correspondences between the two checkpoints (Section 4.1), which serve as a diagnostic when the verdict is **RELATED**.

Lineage follows initialization, not behavior. Under this definition, what determines lineage is where the weights came from, not what the model does. A distilled student trained from freshly initialized weights is unrelated to its teacher, no matter how closely it imitates the teacher’s outputs; model-extraction copies that replicate a model’s function through query access (Tramèr et al. 2016) are likewise unrelated. Conversely, a student initialized from its teacher’s weights is weight-related regardless of the distillation objective: initialization, not training procedure, determines lineage. The method is thus a forensic primitive for weight-level provenance, complementing cryptographic signing, metadata, and watermarking where those were never applied; it is not a legal ownership proof.

2.2 Related Work

We organize prior work by the signal each method measures; Table 1 summarizes the comparison.

Cryptographic and metadata provenance. Cryptographic hashes verify exact byte-level identity but fail af-

ter any weight modification: the avalanche property ensures that fine-tuning, pruning, or even floating-point rounding produces an entirely different digest. Model cards (Mitchell et al. 2019) and experiment-tracking systems document provenance but require honest distributors and preserved records; they can be falsified or lost when checkpoints are redistributed outside controlled environments. Proof-of-learning protocols (Jia et al. 2021) verify that a claimed party performed the training, but require retained training transcripts: forethought that already-released checkpoints lack.

Embedded ownership signals. Parameter watermarks (Uchida et al. 2017) and backdoor watermarks (Adi et al. 2018) embed deliberate signals during training for later extraction. Lukas et al. (2022) found no surveyed scheme reliably robust against fine-tuning and pruning within realistic compute budgets, and all require proactive insertion: unwatermarked legacy checkpoints cannot be verified retroactively. Output watermarks like SynthID mark AI-generated content, not weights: a different problem.

Behavioral and decision-boundary fingerprints. IP-Guard (Cao, Jia, and Gong 2021) extracts inputs near the classification boundary as fingerprints; adversarial frontier stitching (Le Merrer, Perez, and Trédan 2020) uses adversarial examples to mark decision surfaces. These measure functional similarity and cannot distinguish weight inheritance from imitation: a distilled student may share decision surfaces while having entirely independent weights.

Representation similarity. CKA (Kornblith et al. 2019) and SVCCA (Raghu et al. 2017) compare learned representations using activation patterns on a probe dataset. They may regard a successful distilled student as similar: reasonable for their objective (representation alignment) but not ours (weight ancestry). They also require data and forward passes, whereas our method operates on weights alone.

Weight-space provenance. Generic parameter-space similarity (weight cosine, Frobenius distance, permutation-aligned matching (Ainsworth, Hayase, and Srinivasa 2023)) can detect derived checkpoints when the suspect remains close to the reference in parameter space. But these are uncalibrated global scalars: they localize nothing, offer no account of why proximity implies lineage or when it stops doing so, and their margins are not grounded in any property of training. A recent line builds purpose-built provenance on weights, but each requires something beyond the

two checkpoints: HuRef (Zeng et al. 2025) relies on convergence of LLM parameter directions after pretraining; REEF (Zhang et al. 2024) and neural lineage detection (Yu and Wang 2024) need probe data, activations, or gradients; MoTHER (Horwitz, Shul, and Hoshen 2025) recovers directed model trees but requires a collection of checkpoints rather than a single pair. None of these can say which weights were inherited. Our fingerprint is read block by block from trained residual structure: it is data-free, compares a single pair of checkpoints against a calibrated null, and reveals not only whether a suspect inherited weights but which blocks it inherited and how many.

3 Training-Induced Diagonal Dominance in Residual Branch Products

A residual block computes $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$, where F_ℓ is the residual branch. For a branch with K linear layers $W_1, \dots, W_K$ interleaved with nonlinearities, we define the *residual branch product* by composing the linear factors and dropping the nonlinearities between them:

$$M_\ell = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \quad (1)$$

Individual factors W_k are rectangular and do not share an input–output basis; their composed product maps the residual stream back into its own coordinate space, so its diagonal entries compare like with like. The exact factorization is architecture-dependent (Table 2).

3.1 Why Residual Training Induces Diagonal Dominance

Under the Frobenius inner product $\langle A, B \rangle = \text{tr}(A^\top B)$, the identity matrix and the traceless matrices span orthogonal subspaces of $\mathbb{R}^{d \times d}$. Any branch product therefore splits into a component along the identity, with coefficient $\langle M_\ell, I \rangle / \langle I, I \rangle = \text{tr}(M_\ell)/d$, and an orthogonal traceless remainder:

$$M_\ell = -\varepsilon_\ell I + E_\ell \quad (2)$$

where $\varepsilon_\ell = -\text{tr}(M_\ell)/d$ and $\text{tr}(E_\ell) = 0$. The sign convention absorbs the empirical regularity that traces of trained branches are predominantly negative (Section 5), making ε_ℓ typically positive; nothing downstream depends on the sign, since the score below uses the trace magnitude. The decomposition itself is a coordinate choice, available for any matrix.

We measure this concentration with the *diagonal-dominance score*

$$s(M) = \frac{|\text{tr}(M)|}{\|M\|_F}, \quad (3)$$

the fraction of the matrix’s energy lying along the identity direction, up to a factor of $\sqrt{d}$. The finding is that training moves energy into the identity component: in trained residual networks, correctly paired branch products score far above unpaired or untrained ones, a concentration absent at initialization and never developed without skip connections (Figure 1b).

The skip connection explains why. It places the learned branch in an identity-centered parameterization: training optimizes a correction around an explicit identity path rather than learning the full transformation from scratch. For stable gradient flow, the block Jacobian $J = I + J_F$ should remain near-orthogonal throughout training, a condition known as dynamical isometry (Pennington, Schoenholz, and Ganguli 2017). If $J \approx I$, then $J_F \approx 0$; for small but nonzero learned corrections, the linear component of J_F tends toward $-\varepsilon I$ with small ε . Because W_{in} and W_{out} receive coupled gradients through the branch, this pressure aligns their product with coordinate-preserving directions. Non-residual networks lack this constraint: without the skip connection, there is no functional pressure toward identity, and nothing couples W_{in} with W_{out} more than with any other weight matrix.

This is a mechanistic hypothesis, not a theorem, but each of its predictions is testable. It predicts no signal at initialization under any scheme: an ablation over seven initializations finds chance-level pairing before training, including orthogonal initialization, which satisfies dynamical isometry by construction yet has learned nothing (Saxe, McClelland, and Ganguli 2014) (Appendix A.1). It predicts no signal without the skip connection: a PlainNet trained to comparable loss develops none (Appendix A.2).

3.2 Margin Analysis

For the score to separate correct pairings from incorrect ones, two things must hold: correctly paired products must score high, and mismatched products must score low. We treat each in turn.

By the orthogonality of decomposition (2), $\|M_\ell\|_F^2 = \varepsilon_\ell^2 d + \|E_\ell\|_F^2$: the branch product’s energy splits exactly between its identity component and its traceless remainder. The score therefore depends on M only through the ratio of the two:

Proposition 1 (Correct-pair score). *Let $M = -\varepsilon I + E$ with $\text{tr}(E) = 0$ and $\varepsilon \neq 0$. Then:*

$$s(M) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \quad (4)$$

with equality if and only if $E = 0$.

Proof. $|\text{tr}(M)| = |\varepsilon|d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ by orthogonality. Substitute into (3).

The score is $\sqrt{d}$ discounted by how much the learned residue E outweighs the identity component: geometrically, $s(M)/\sqrt{d}$ is the cosine of the angle between M and the identity. A correct pair scores high exactly to the extent that training concentrated its product along the diagonal.

A mismatched product has no such concentration. For W_{out} and W_{in} drawn from different blocks, the product has no preferred alignment with the identity: its trace is a sum of d weakly correlated terms of comparable size to the other d^2 entries, giving

Corollary 1 (Mismatched baseline). *For products of projections from independent blocks, $\mathbb{E}[s] \approx 1/\sqrt{d}$.*

The margin between the two regimes therefore scales as d : wider networks separate correct from incorrect pairings more cleanly, not less. Empirically the correct-pair score grows as $d^{0.87}$ (Section 5), faster than the $\sqrt{d}$ envelope’s rate, consistent with $\|E\|_F/(\varepsilon\sqrt{d})$ shrinking as d grows, while always remaining below the $\sqrt{d}$ bound. The proposition bounds the score; it does not fix its growth rate.

4 From Diagonal-Dominance Fingerprints to Lineage Verification

Diagonal dominance identifies correctly paired trained residual branches, but it does not establish lineage: unrelated trained models both exhibit it. The decomposition (2) supplies the division of labor. The identity component is generic; its strength certifies that a branch carries trained residual structure, and Section 5.2 uses it to recover projection pairings within a model. But precisely because it is shared across trained models, it says nothing about ancestry. The traceless component E_ℓ is block- and checkpoint-specific: it survives weight-preserving transformations but not reinitialization. Verification therefore runs on the traceless component, with diagonal dominance serving as a gate on which branches are trustworthy. This section builds the protocol in three steps: signatures, alignment, decision.

4.1 Centered Residual Signatures and Block Alignment

For each branch product M_ℓ we retain only the traceless remainder of decomposition (2), normalized to a unit vector:

$$R_\ell = M_\ell - \frac{\text{tr}(M_\ell)}{d} \cdot I, \quad \phi_\ell = \frac{\text{vec}(R_\ell)}{\|\text{vec}(R_\ell)\|_2} \quad (5)$$

R_ℓ is exactly the E_ℓ of (2); the signature ϕ_ℓ carries the block- and checkpoint-specific geometry that weight-preserving transformations inherit and independent training cannot reproduce. Because M_ℓ is exactly invariant under the two function-preserving reparameterizations of a residual branch (hidden permutation: $W_{\text{in}} \leftarrow PW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}P^\top$; reciprocal rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}/c$), so is every signature built on it: re-basing hidden units cannot evade the comparison (Section 6).

Given reference A and suspect B , each with L blocks, we form the signature-similarity matrix $G_{ij} = \langle \phi_i^A, \phi_j^B \rangle$, gated so that branches with weak diagonal dominance in either checkpoint contribute conservatively (Appendix C), and recover the block correspondence as the assignment maximizing total similarity,

$$\pi^* = \arg \max_{\pi \in \mathcal{P}_L} \sum_i G_{i, \pi(i)}, \quad (6)$$

via the Hungarian algorithm (Kuhn 1955). For derived checkpoints that preserve block ordering, π^* should be the identity permutation, and recovering it without metadata is evidence that the signatures track block identity rather than position; the Hungarian step matters when ordering cannot be trusted, and its output doubles as a diagnostic: when the verdict is RELATED, the correspondence matrix shows which blocks were inherited (Appendix A). Computing the

branch products costs $O(Ld^2h)$, where h is the branch’s inner width; the similarity matrix adds $O(L^2d^2)$ and the assignment a negligible $O(L^3)$.

4.2 Lineage Score and Verification Test

The lineage score averages signature similarity over aligned branches:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^L \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \quad (7)$$

Each term is a cosine, so $\mathcal{L} \in [-1, 1]$ with $\mathcal{L}(A, A) = 1$. The score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$: exchanging the roles of A and B transposes G , and the optimal assignment value is invariant under transposition, provided the gate is defined over both checkpoints jointly (Appendix C). Related checkpoints should score near 1 and unrelated ones near 0, since independently initialized weights produce uncorrelated E_ℓ ; Section 5 tests both predictions.

To map the score onto the verdicts of Section 2, we calibrate against a null distribution: with $\mathcal{N}_A = \{\mathcal{L}(A, U_j)\}$ the scores between A and independently trained models U_j , and (μ_0, σ_0) its mean and standard deviation,

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_0}{\sigma_0} \quad (8)$$

The protocol returns

$$V(A, B) = \begin{cases} \text{RELATED} & \text{if } Z > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (9)$$

with symmetric thresholds $\tau_{\text{upper}} = 3.0$, $\tau_{\text{lower}} = -3.0$, giving a false-positive rate below 0.13% under a Gaussian null. Compatibility (same layer count L and hidden dimension d , per the setting of Section 2) is a precondition checked before any scoring; the protocol returns INCOMPATIBLE when it fails, an enforcement of the setting rather than a fourth verdict. The verifier needs weights alone, and $O(L^2 + d^2)$ working memory beyond the checkpoints themselves.

5 Experiments

Our experiments validate the claim ladder of Sections 3–4 in order. Section 5.1 tests the three mechanistic predictions of Section 3.1: the fingerprint is absent at initialization under every scheme tested, never develops without skip connections, and emerges within the first few epochs of residual training. Section 5.2 tests the extraction machinery of Section 4.1 and the margin analysis of Proposition 1, showing that architecture-aware branch products recover block correspondences across 12 architectures spanning language, vision, and speech, with a correct-pair score that grows as $d^{0.87}$ against the $\sqrt{d}$ envelope. Section 5.3 tests the full verification protocol of Section 4.2 at three levels of realism: controlled MLP benchmarks with known ground truth, a CIFAR-10 ResNet-18 benchmark, and real fine-tuned derivatives of LLaMA-2-7B scored against a cross-model null. Section 5.4 positions the method against weight-space, activation-based, and decision-boundary baselines. Extended ablations are in Appendices A and D.

Model	Layers	Pair Acc	AUC
GPT-2 (124M–1.5B)	12–48	100%	1.00
BERT-base / ViT-B/16	12	100%	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	100%	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	100%	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	80–100%	0.93–1.00
Whisper (tiny/base/small)	4–12	100%	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 3: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Sub-100% SwiGLU sub-paths are rescued by joint factorization. Random-init baselines: $\leq 9\%$.

5.1 The Fingerprint Exists and Is Learned

Section 3.1 predicts that diagonal dominance is learned, not an initialization artifact, and requires the skip connection. We test each prediction.

Emergence during training. On a 48-block residual network, block-pairing accuracy is at the chance baseline (6.25%) at epoch 0 and reaches 100% by epoch 5, well before loss plateaus. By epoch 300, the mean correct-pair score exceeds the mean incorrect-pair score by $21\times$ (Appendix A). The fingerprint develops rapidly in early training.

The skip connection is necessary. We train a PlainNet—architecturally identical to ResNet except that skip connections are removed—to comparable evaluation loss (1.56 vs 1.35). On a 24-block network (chance $\approx 4.2\%$), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace; PlainNet achieves only 3% accuracy (at chance) with 47% negative-trace incidence and AUC 0.51 (Appendix A.2). The skip connection is structurally necessary.

Not an initialization artifact. An ablation over seven initialization schemes—Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal—finds $\leq 2.1\%$ pair accuracy before training across all schemes (Appendix A.1). After training, six schemes reach 94–100% accuracy. Orthogonal initialization is the critical case: it provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), satisfying the near-identity Jacobian condition of Section 3.1, yet shows *zero* correct pairs before training. This rules out the alternative explanation that near-orthogonal Jacobians alone create the fingerprint; what matters is isometry under training pressure. The seventh case, $\sigma=0.02$ Gaussian, fails (13% trained) because blocks collapse to near-zero contribution, removing the Jacobian constraint.

5.2 The Fingerprint Generalizes and Scales

Table 3 reports block-pairing accuracy across 12 architectures spanning language (GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1), vision (ViT, ResNet), and speech (Whisper): 91–100% in every family, against random-initialization baselines of at most 9%. The same extraction transfers across GELU and SwiGLU activations and across causal, bidirectional, and grouped-query attention with no

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
LoRA merge	0.98	0.97	$\sim$
Independent / Distilled	0.08	0.01	-

Table 4: Lineage score survival under post-training modification. All 75 related checkpoints detected at 1% FPR; unrelated checkpoints stay below 0.20. The LoRA row reports preliminary results ($\sim$); formal detection counts pending.

per-family tuning; what changes between families is only the factorization of Table 2.

Factorization must match the architecture. The branch product carries the signal only when it composes every linear factor of one branch. For ResNet Bottleneck blocks the two-factor product W_3W_1 pairs at chance, while the full $W_3W_2W_1$ recovers 91–100% (Appendix D); for SwiGLU MLPs, scoring the gate and up paths separately loses accuracy that the joint product $W_{\text{down}}W_{\text{up}}$ restores. Incomplete products do not weaken the fingerprint—they miss it, which is itself informative: the structure lives in the composed map from residual stream to residual stream, not in any individual weight matrix.

RLHF and behavioral pedigree. RLHF leaves the fingerprint intact—LLaMA-2 base to chat pairs at 100%—while the Llama-3-derived DeepSeek distill is correctly rejected (Section 5.3). Pairing follows the weights, not the training story.

Scaling with hidden dimension. The mean correct-pair score grows from 4.18 at $d=768$ (GPT-2) to 23.83 at $d=5120$ (LLaMA-2-13B), an empirical $d^{0.87}$ that remains within the $\sqrt{d}$ envelope of Proposition 1 (observed scores reach 15–19% of the bound). Corollary 1 places the mismatched baseline at $1/\sqrt{d}$, so separation compounds from both directions as networks widen: the correct-to-incorrect ratio improves from $35\times$ on GPT-2 to $2,140\times$ on LLaMA-2-13B, and holds at $1,530\times$ across the 256 experts of Mixtral-8x7B. Scale works for the verifier.

5.3 From Fingerprints to Lineage Verification

Diagonal dominance certifies trained residual structure, not ancestry: unrelated trained models exhibit it equally, and a verifier built on the dominance score alone achieves AUROC=0.417 on the MLP benchmark (Figure 3a)—the failure Section 4 predicts, and the reason verification runs on the centered signatures ϕ_ℓ . We test the protocol on controlled MLPs with known ground truth, then a trained-from-scratch ResNet benchmark, then the public LLaMA-2 family.

Controlled MLP benchmark. From 2 trained references we construct 52 pairs: five related-checkpoint kinds (fine-tuning, fine-tuning to a different target, Gaussian noise, magnitude pruning, quantization) against different-seed and distilled negatives; the expanded 159-pair set draws on a 24-checkpoint zoo. Related ($n=75$) and unrelated ($n=84$) checkpoints separate completely (Figure 1c), giving

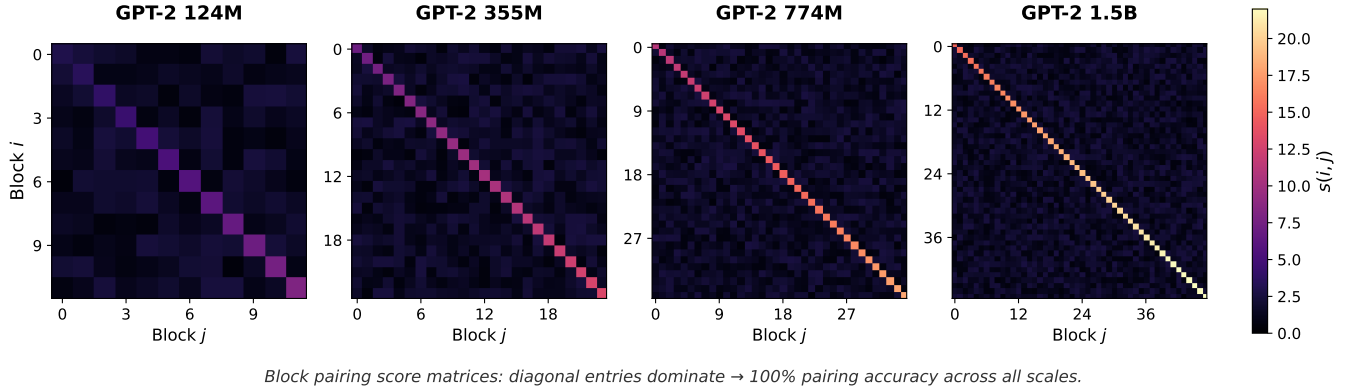


Figure 2: Block pairing score matrices $s(i, j)$ for GPT-2 models from 124M to 1.5B parameters. Diagonal entries dominate, yielding 100% pairing accuracy across all scales. Trace analysis and method comparisons in Appendix D.1.

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
<i>Related (real fine-tunes)</i>				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
<i>Unrelated (cross-model null)</i>				
Mistral-7B-v0.1	scratch	4e−4	0.158	UNR.
OpenLLaMA-7B	scratch [†]	4e−4	0.089	UNR.
Yi-6B	scratch	5e−4	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	4e−4	0.162	UNR.

Table 5: Real-LLM lineage validation on the LLaMA-2 family (7 pairs). [†]OpenLLaMA shares LLaMA-2’s exact architecture ($d=4096$, $L=32$) but is trained from scratch; its classification as unrelated cannot come from configuration.

AUROC=1.000 and 75/75 detection at 1% FPR (Table 4). The margin is not thin at its weakest point: the 85%-pruned checkpoint ($\mathcal{L}=0.58$) still scores $3\times$ above the strongest unrelated one, a distilled student at $\mathcal{L}=0.19$.

Real architecture, controlled lineage. On CIFAR-10 (Krizhevsky 2009) ResNet-18 the same protocol gives AUROC=1.000 with TPR@1%FPR=100%: related checkpoints score $\mathcal{L} \in [0.695, 1.000]$ while the strongest unrelated checkpoint reaches only 0.004, more than $100\times$ below the weakest related one (Appendix A).

Public models, real lineage. With LLaMA-2-7B as reference, all three real fine-tuned variants are flagged: the RLHF chat model ($\mathcal{L} = 0.995$), the Vicuna instruction fine-tune ($\mathcal{L} = 0.996$), and CodeLlama ($\mathcal{L} = 0.336$), whose heavy code-specialized continued pretraining makes it the weakest relative—yet it remains $\sim 700\times$ above the null. The four independently trained negatives score $\mathcal{L} \in [3.5, 4.8] \times 10^{-4}$, defining a real cross-model null ($\mu_0 = 4.2 \times 10^{-4}$). OpenLLaMA is the decisive case among them: it reproduces LLaMA-2’s exact shape ($d=4096$, $L=32$, $d_{\text{ff}}=11008$) from scratch, so nothing in its configuration distinguishes it from a true descendant—the UNRELATED ver-

dict can only come from the weights. Mean separation is $\approx 1850\times$, against $6\times$ for weight cosine; aligned Frobenius and singular-value distance underperform substantially on this set (AUROC=0.667). With four independents this is a descriptive separation check rather than a calibrated false-positive rate (Appendix D.4).

Beyond binary verdicts. On a branching tree ($C_1 \rightarrow \{C_2, C_3\}$, $C_2 \rightarrow C_4$, $C_3 \rightarrow C_5$), $\mathcal{L}$ ranks C_4 ’s relatives in genealogical order: parent (0.96) > grandparent (0.91) > uncle (0.87) > cousin (0.85) $\gg$ independents (≈ 0.08) (Appendix C). The score does not merely cross a threshold; it decays with distance from the ancestor—which is what ancestry recovery beyond a binary verdict would require.

5.4 Comparison with Data-Free and Behavioral Baselines

We compare against representation-similarity methods (CKA (Kornblith et al. 2019), SVCCA (Raghu et al. 2017)), weight-space baselines (aligned Frobenius, singular-value distance, weight cosine), and a decision-boundary baseline (IPGuard (Cao, Jia, and Gong 2021)) on the 52-pair MLP benchmark (Table 6).

The saturated benchmark. Four data-free weight-space methods reach AUROC=1.000, so the headline number cannot adjudicate between them. The Gap-Z column and per-kind breakdown (Appendix D.3) carry the real signal: IPGuard ranks related checkpoints *below* distilled ones (Gap-Z = −3.5); CKA’s Gap-Z = +8.6 is large in σ units only because σ_{distill} is tiny (0.014), while the raw ratio ($0.999/0.827 \approx 1.21\times$) is fragile.

The contribution claim. We do not claim that diagonal dominance beats weight cosine on aggregate AUROC. The contribution is that the residual-product signature (i) yields per-block correspondence as a by-product, (ii) admits the closed-form margin analysis of Proposition 1, and (iii) is interpretable in terms of trained residual geometry. On the real 7B weights of Section 5.3, the $1850\times$ -vs- $6\times$ margin gap matters: aligned Frobenius and SV-distance underperform substantially on this real-model set (AUROC=0.667, above chance but far from the saturation reached on syn-

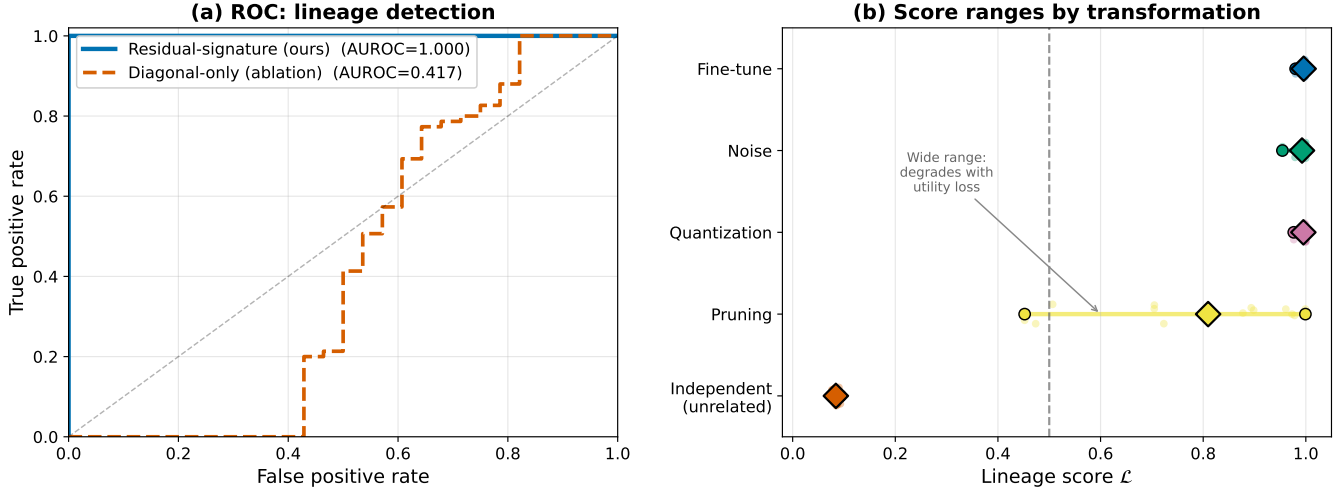


Figure 3: Lineage verification on depth-24 residual MLPs. (a) ROC comparison: the residual-signature score achieves AUROC=1.000, while diagonal-dominance alone fails (AUROC=0.417)—it cannot distinguish two trained residual models. (b) Score ranges by transformation type: fine-tuning, noise, and quantization preserve the fingerprint near 1.0; independent (unrelated) models score below 0.1; only pruning shows a wide range, degrading as model utility degrades ($\rho = -0.83$).

Method	Type	Data-free	AUROC	Gap-Z $\uparrow$	Block Corr.
Diagonal Dominance (ours)	Weight-space	✓	1.000	+53.0	✓
Aligned Frobenius	Weight-space	✓	1.000	+72.0	✓
Singular Value Distance	Weight-space	✓	1.000	+7.9	✗
Weight Cosine	Weight-space	✓	1.000	+76.3	✗
SVCCA (Raghu et al. 2017)	Activation-space	✗	1.000	+45.4	✗
CKA (Kornblith et al. 2019)	Activation-space	✗	0.829	+8.6	✗
IPGuard (Cao, Jia, and Gong 2021) (regression analogue)	Decision-boundary	✗	0.500	-3.5	✗

Table 6: Strong-baseline comparison on the MLP lineage benchmark (52 pairs). Four data-free weight-space methods reach AUROC=1.000; Gap-Z measures standardized distillation rejection, where IPGuard’s -3.5 indicates related checkpoints rank *below* distilled. Block Corr.: only diagonal dominance and aligned Frobenius provide per-block correspondence.

thetic benchmarks).

The harder regime. On layer grafts and linear merges (Appendix D.5, 40 pairs), diagonal dominance, aligned Frobenius, and weight cosine remain at or near AUROC=1.000. Singular-value distance underperforms on merges (0.448); CKA drops to ≈ 0.80 ; IPGuard achieves AUROC=0.05 on grafts—below chance, indicating systematic misclassification where its decision-boundary assumptions break down. The diagonal-dominance score retains Spearman $\rho \approx 0.96$ with the reference fraction, supporting the interpretability claim: it tracks partial overlap monotonically, not merely crossing a threshold.

6 Robustness and Boundary Cases

Section 5 established that the protocol detects lineage. This section maps its edges under three pressures: benign post-training modification (Section 6.1), deliberate evasion (Section 6.2), and definitional boundaries where the method correctly declines or genuinely fails (Section 6.3).

6.1 Survival Under Post-Training Modification

The question is not whether the fingerprint detects lineage—Section 5.3 established that—but what happens as transformation intensity increases. Table 4 shows survival across five transformation families: fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), quantization (16–256 levels), and LoRA adapter merging.

Fine-tuning, quantization, and noise preserve the signal with mean $\mathcal{L} = 0.99$ and minimum 0.97. Fine-tuning to a different target degrades to mean 0.94 (min 0.84). Pruning shows the steepest decline: from $\mathcal{L} = 0.99$ at 10% sparsity to 0.58 at 85% sparsity. LoRA-merged checkpoints score $\mathcal{L} \in [0.97, 0.99]$ (~formal detection counts pending). CodeLlama ($\mathcal{L} = 0.336$), the real-world far endpoint of the fine-tuning axis after heavy code-specialized continued pre-training, remains $\sim 700\times$ above the unrelated null.

The thesis: fingerprint degradation correlates with utility loss ($\rho = -0.83$; Figure 3b). No benign path removes the signal without damaging the model. The two are structurally coupled through the Jacobian constraint of Section 3.1.

Condition	Verdict
Different architectures	INCOMPATIBLE
Non-residual networks	NO SIGNAL
Distillation (cross-arch)	OUT OF SCOPE
Training data overlap	NOT DETECTED
Two indep. same-arch	CANNOT DISTINGUISH

Table 7: What the method does NOT do.

6.2 Adversarial Suppression and Function-Preserving Symmetries

Function-preserving symmetries. Per-block permutation ($W_{\text{in}} \leftarrow PW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}P^\top$) and reciprocal rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ to numerical precision; orthogonal rotations also preserve M but break the network, since ReLU does not commute with rotation. Re-basing hidden units therefore buys an adversary nothing: the signature is built on the one object these reparameterizations cannot move.

Gradient-based suppression. A stronger adversary optimizes a perturbation Δ directly against the score while constraining function:

$$\min_{\Delta} \mathcal{L}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (10)$$

Sweeping λ traces a Pareto frontier (Appendix E). With $\lambda \geq 1$ the attack cannot push $\mathcal{L}$ below 0.91; reaching the unrelated range ($\mathcal{L} \approx 0.08$) costs at least 12% eval-loss degradation; removing the constraint destroys the model (a $57\times$ loss increase). An attacker with white-box knowledge of the score faces the same trade that Section 6.1 measured for benign transformations: the signal does not come off without taking function with it.

Untested attack surfaces. We have evaluated one adaptive attack; two remain open. Re-alignment toward a third model in the style of Git Re-Basin could attempt to rotate the E matrices off their checkpoint-specific directions, and fine-tuning with an explicit decorrelation penalty on $\langle E_{\text{suspect}}, E_{\text{reference}} \rangle$ is a fingerprint-aware strengthening of the attack above; we leave both to future work. Warm-start distillation—a student initialized from teacher weights, then distilled—is not an evasion candidate: under the definition of Section 2 it is RELATED, and it functions as a positive control on the distillation boundary rather than an attack on it.

6.3 Boundary Cases and Failure Modes

Some conditions the method handles correctly but uninformatively; others it cannot handle at all. We separate the two.

Correct boundary behaviors. Distillation is the most important. Students trained from fresh weights are UNRELATED by definition, and the verdicts agree: same-architecture distilled students score $\mathcal{L} = 0.086$ on the controlled benchmark, and the DeepSeek distill sits at the 10^{-4} cross-model null on the real test (Sections 5.3–5.4). The gap between those two magnitudes is itself informative—same-architecture distillation on shared data retains a trace of ac-

cidental correlation that cross-family comparison lacks—and both fall well below threshold. Symmetry is the second boundary: $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol cannot name the ancestor; direction requires metadata or a third candidate, as in the branching tree of Section 5.3. Partial lineage is the third: grafts and merges form a graded regime in which $\mathcal{L}$ tracks the fraction of inherited blocks with Spearman $\rho \approx 0.96$ (Appendix D.5)—partial overlap is measured, not misclassified.

Genuine failures (Table 7). The protocol returns INCOMPATIBLE when depth or width differ, and there is no signal to read in non-residual networks (the PlainNet of Section 5.1) or when blocks collapse to zero contribution (the $\sigma = 0.02$ initialization). Utility-destroying perturbations such as 2-bit quantization take the fingerprint down with the model, and no verdict is possible about a common ancestor that neither party can produce.

The INCONCLUSIVE verdict. Across all 280+ tested pairs, every score fell cleanly above τ_{upper} or below τ_{lower} ; no verdict was INCONCLUSIVE. The band exists for deployments our benchmarks do not cover—nulls built from few independents, or suspects near the graft and merge mid-points of Appendix D.5—where scores need not separate as sharply as they do here.

7 Discussion

Scope. The method requires white-box access and residual architecture. Non-residual networks lack the identity-target constraint that creates the fingerprint.

Positioning. This complements watermarking: watermarks require forethought; diagonal dominance offers passive, retroactive verification for checkpoints never watermarked.

Validation scope. Lineage AUROC is reported on the synthetic MLP zoo and CIFAR-10 ResNet-18, with real-architecture evidence from pairing on Transformer families and a 7-pair LLaMA-2 test. The remaining gap is tree breadth: all related checkpoints derive from a single base. Extending to additional checkpoint trees and larger negative sets is straightforward engineering. Detailed limitations are in Appendix G.

8 Conclusion

Training leaves traces. Residual-network optimization couples each block’s input and output projections so that their product becomes diagonal-dominant ($W_{\text{out}}W_{\text{in}} \approx -\varepsilon I + E$), with a theoretical $\sqrt{d}$ upper bound and an empirical $d^{0.87}$ growth observed from GPT-2 through LLaMA-2-13B and Mixtral-8x7B. Across 12 architectures spanning 5 families, the fingerprint recovers 91–100% of block correspondences and achieves AUROC=1.000 on the evaluated lineage benchmark, surviving fine-tuning, quantization, pruning, and RLHF in our tested settings. Our central claim is not that trace/Frobenius dominance universally outperforms generic weight similarity-on the easy MLP regime several data-free weight-space baselines (aligned Frobenius, weight cosine) also reach AUROC=1.000—but that trained residual products expose an interpretable, block-localized,

data-free provenance signal that supports both correspondence recovery and lineage verification, and that this signal is substantially more robust to distillation than activation- and decision-boundary baselines. Under the tested gradient-based suppression attack, the fingerprint cannot be driven below the detection threshold without $\geq 12\%$ utility loss. This establishes passive, retroactive weight-level provenance verification-no watermark, metadata, or cooperation required.

Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 4, 6) is constructed from 2 trained reference checkpoints, each paired with 5 related-checkpoint kinds $\times 6$ derivative seeds = 30 truly-related pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as unrelated checkpoints. The expanded 159-pair set used in Figure 3 draws from the full 24-checkpoint MLP zoo with the same transformation grid. Exact parameters are listed in Appendix A; null calibration uses the 50 different-seed pairs from Appendix C. Real-architecture pairing (Table 3) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IPGuard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

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A Extended Experimental Details

A.1 Initialization Ablation

We test whether the diagonal-dominance fingerprint depends on initialization scheme. We train depth-24 residual MLPs ($h=64$, $d=24$) on a synthetic regression target for 200 epochs from seven initialization schemes: Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal. Table 8 reports pair accuracy before and after training.

Init scheme	Untrained	Trained	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian $\sigma=0.02$	2.1%	13%	0.671

Table 8: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq 2.1\%$) at initialization, rising to 75–100% after training. Orthogonal initialization—which provides dynamical isometry at init—shows *zero* correct pairs before training.

Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The $\sigma=0.02$ small-Gaussian case fails because residual blocks collapse to near-zero contribution; without active per-block contribution there is no Jacobian constraint to enforce.

A.2 PlainNet Control Experiment

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical to ResNet except that skip connections are removed.

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Table 9: ResNet vs. PlainNet control. Despite similar evaluation loss, PlainNet achieves only chance-level pair accuracy (3% vs. chance $\approx 4.2\%$), confirming the skip connection is necessary for the fingerprint.

While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline—with AUC 0.51 and no discernible diagonal structure. The

skip connection creates a functional constraint that imprints the characteristic trace structure.

A.3 Training Dynamics

The fingerprint emerges rapidly during training. On a 48-block residual network:

- Epoch 0: Mean correct-pair score = 0.19, pair accuracy = 6.25%
- Epoch 5: Diagonal fully separated, pair accuracy = 100%
- Epoch 300: Mean correct-pair score = 2.07, mean incorrect-pair score = 0.16 ($21\times$ ratio)

The transformation occurs within the first few epochs, well before training loss plateaus.

B Extended Theoretical Results

B.1 Null Model (Corollary)

Corollary 2 (Null Model). *For randomly initialized weights with no training, all pairs satisfy $\mathbb{E}[s(i, j)] = 1/\sqrt{d} + o(1)$ uniformly in (i, j) . Hungarian matching recovers correct pairs at the chance rate $1/L$.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

B.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\text{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score *well above* the random baseline $\Theta(1/\sqrt{d})$, with empirical growth of approximately $d^{0.87}$ across $d \in [768, 5120]$ (Appendix D), bounded above by $\sqrt{d}$ but typically reaching only 15–25% of that bound. The signal-to-baseline ratio therefore grows roughly as $d^{1.37}$, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i, i) \in [1.76, 3.23]$ —well above the baseline but below the theoretical maximum of $\sqrt{d}$.

C Verification Protocol Details

C.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right) \quad (11)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

C.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \quad (12)$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

C.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}, \{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}, \{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{RELATED} & \text{if } Z(A, B) > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z(A, B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (13)$$

C.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

C.5 Lineage Chain and Branching Tree

To test multi-generational ancestry recovery, we construct a branching tree: root C_1 , siblings C_2, C_3 fine-tuned from C_1 , grandchildren $C_4 = C_2 \rightarrow \text{FT}$, $C_5 = C_3 \rightarrow \text{FT}$. The lineage score recovers correct ancestry: for C_4 , ranking by $\mathcal{L}(C_4, \cdot)$ is parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $>$ independent (0.08). The score decays monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

D Per-Architecture Details

D.1 GPT-2 Scaling and Trace Analysis

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: $d=768$ at $\bar{s}(i, i) = 4.18$, $d=1024$ at 5.46, $d=1280$ at 6.98, $d=1600$ at 7.77. The fitted exponent within the GPT-2 family is ≈ 0.85 , consistent with the cross-family $s \propto d^{0.87}$ scaling reported in the main text; the $\sqrt{d}$ upper bound itself is never violated (observed scores reach 15–19% of the bound).

Figure 4 shows the trace values $\text{tr}(W_{\text{out}}W_{\text{in}})$ per block across all four GPT-2 variants. Negative traces (blue) constitute 81–92% of blocks, confirming the diagonal-dominance mechanistic prediction. The early blocks (0–2) consistently

show positive traces, likely reflecting the different role of embedding-adjacent layers.

D.2 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Figure 5 shows the scaling behavior: Frobenius norm matching degrades from 67% (GPT-2 124M) to 52% (GPT-2 1.5B), while diagonal dominance remains at 100%. Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

D.3 Per-Kind Lineage Baseline Breakdown

Table 10 reports the mean lineage score of each baseline broken down by related-checkpoint kind (fine-tune, fine-tune to a new target, noise, prune, quantize) and unrelated-checkpoint kind (different-seed same-task, distilled). The related rows should be high; the unrelated rows should be low. The aggregate AUROC reported in Table 6 hides two structural failures that this breakdown exposes:

- **CKA and SVCCA fail on distilled unrelated checkpoints.** CKA scores distilled students at 0.83 and SVCCA at 0.89—only $1.2\times$ and $1.13\times$ below the corresponding fine-tuned related checkpoints (0.999 and 0.9997). On the present benchmark there are 16 different-seed unrelated checkpoints and only 6 distilled, so AUROC remains high; any reweighting toward distillation would collapse it.
- **IPGuard scores noise-perturbed related checkpoints at 0.000.** The decision-boundary analogue ranks all 6 noise-perturbed related checkpoints below all unrelated ones, driving overall AUROC to chance (0.500).

All four weight-space methods (diagonal dominance, aligned Frobenius, singular-value distance, weight cosine) score distilled unrelated checkpoints at or below the different-seed baseline—they cleanly reject distillation, while the activation- and decision-boundary methods cannot.

D.4 Real-LLM Lineage Validation

To move beyond block-correspondence pairing on real foundation models, we evaluate $\mathcal{L}(\cdot, \cdot)$ on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.4. Using LLaMA-2-7B as reference, we score seven real reference-vs-model pairs: three real fine-tuned *related* checkpoints and four independently-trained *unrelated* checkpoints. For each model we extract the per-layer SwiGLU branch product $M_\ell = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times d}$, form the centered residual signature ϕ_ℓ , and score $\mathcal{L}(A, B) = \frac{1}{L} \sum_\ell \langle \phi_\ell^A, \phi_\ell^B \rangle$. The same branch products feed the three weight-space baselines (weight cosine, aligned Frobenius, singular-value distance); IPGuard is computed separately from output logits and is available only on the subset of pairs for which logits were collected (2 related, 2 unrelated), and the activation baselines (CKA, SVCCA) are omitted at 7B scale because per-pair computation over 32×32 layers is prohibitively

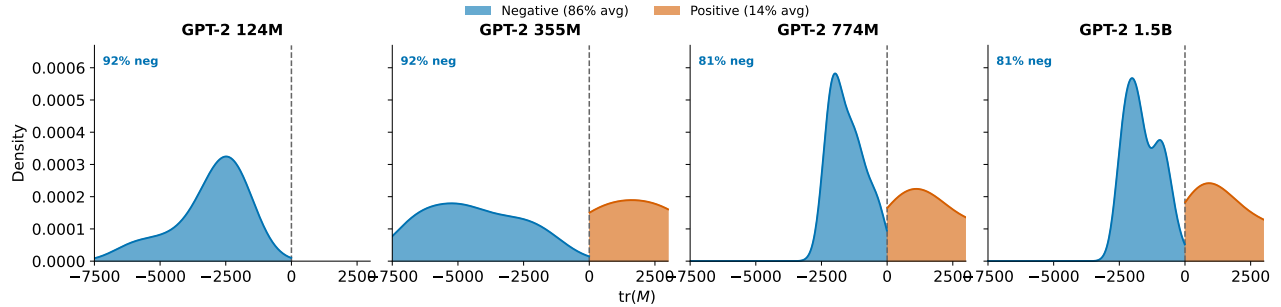


Figure 4: Distribution of trace values $\text{tr}(W_{\text{out}}W_{\text{in}})$ across GPT-2 scales. Blue: negative trace (86% average); orange: positive trace (14% average). The strong skew toward negative values confirms the diagonal-dominance mechanistic prediction.

Method	Related (should be high)					Unrelated (should be low)	
	FT	FT-new tgt	Noise	Prune	Quant	Diff-seed	Distilled
Diag. Dominance (ours)	0.391	0.326	0.400	0.359	0.404	0.029	0.030
Aligned Frobenius	-0.036	-0.301	-0.026	-0.204	-0.007	-1.333	-1.300
Singular Value Dist.	-0.004	-0.023	-0.003	-0.018	-0.001	-0.030	-0.032
Weight Cosine	0.999	0.956	1.000	0.979	1.000	0.096	0.109
SVCCA	1.000	0.953	1.000	0.993	1.000	0.881	0.885
CKA	0.999	0.760	1.000	0.986	1.000	0.820	0.827
IPGuard (regr.)	0.888	0.110	0.000	0.626	0.993	0.632	0.642

Table 10: Per-kind mean lineage scores on the 52-pair MLP benchmark. Each method uses its native scale; what matters is the relative gap between the rightmost two columns (unrelated) and the leftmost five (related). Italicized entries highlight failure modes: SVCCA and CKA give distilled unrelated checkpoints near-related scores; IPGuard scores noise-perturbed related checkpoints at exactly 0.000. All four weight-space methods (including ours) keep distilled below or at the diff-seed baseline. Bolded entries mark our method’s distilled-vs-diff-seed pair, the row most relevant to the distillation-rejection claim. $n=6$ per related kind, $n=16$ diff-seed, $n=6$ distilled.

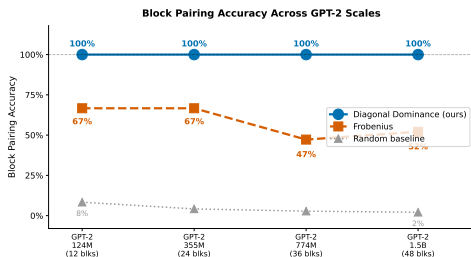


Figure 5: Block pairing accuracy as model size increases. Diagonal dominance (blue) maintains 100% accuracy; Frobenius matching (orange) degrades from 67% to 52%; random baseline (gray) drops from 8% to 2% as the number of blocks increases.

expensive. The benchmark runs on a single GPU via `gpu_benchmark_issue44.py`.

Compatibility. The strict protocol (App. C) returns INCOMPATIBLE when $\text{arch}(A) \neq \text{arch}(B)$. Because M_ℓ contracts the intermediate width d_{ff} and lives in $\mathbb{R}^{d \times d}$, the branch-product comparison itself needs only matching depth L and hidden size d . For the two cross-family negatives

(Mistral and DeepSeek carry $d_{\text{ff}} = 14336$ vs. 11008 for the LLaMA/Yi models) we therefore apply this *relaxed* branch-product compatibility rather than strict architecture equality; we flag it as a deliberate relaxation used only to admit cross-family negatives. Note that OpenLLaMA—the decisive negative—satisfies the strict rule exactly, so the strongest negative does not rely on the relaxation.

Pairs. Related: the official RLHF model Llama-2-7B-chat, the Vicuna-7B instruction fine-tune, and CodeLlama-7B, a code-specialized continued-pretraining derivative of the base. *Unrelated:* four independently-trained checkpoints with no LLaMA-2 weight ancestry—Mistral-7B-v0.1, OpenLLaMA-7B, Yi-6B, and DeepSeek-R1-Distill-Llama-8B. The last is *behaviorally* distilled from DeepSeek-R1 reasoning traces but *weight-derived* from Meta’s Llama-3.1-8B; its UNRELATED verdict w.r.t. LLaMA-2 demonstrates that the method tracks weight inheritance rather than behavioral pedigree, consistent with Section 5.4.

OpenLLaMA: a same-architecture, from-scratch negative. OpenLLaMA-7B is an open reproduction of the LLaMA architecture trained from scratch on different data. It shares LLaMA-2’s exact residual shape ($d = 4096$, $L = 32$, $d_{\text{ff}} = 11008$), tokenizer conventions, and RoPE setup, yet inherits none of its weights. It is therefore the strongest pos-

sible negative: architecture is held *constant* and only weight lineage varies, so a configuration-level check cannot distinguish it from a genuinely related checkpoint—the verdict must come from the weights. Diagonal dominance scores it at $\mathcal{L} = 4.5 \times 10^{-4}$, indistinguishable from the other unrelated checkpoints.

Cross-model null. The four unrelated checkpoints define a real cross-model null $\mu_0 = 4.2 \times 10^{-4}$, $\sigma_0 = 5.0 \times 10^{-5}$ (all four within $[3.5, 4.8] \times 10^{-4}$). With only $n = 4$ real independents this is a *descriptive* separation check rather than a calibrated false-positive-rate estimate; we therefore report the raw $\mathcal{L}$ and the separation in orders of magnitude as the primary evidence, and defer a calibrated FPR to the larger checkpoint trees discussed below. The four unrelated checkpoints span three from-scratch pre-training lineages (Mistral, OpenLLaMA, Yi) plus a Llama-3 derivative, so the null is not dominated by any single family. *Verdict rule for Table 11:* we label a pair RELATED if $\mathcal{L}$ exceeds a descriptive threshold placed an order of magnitude above the null ($\mathcal{L} > 10^{-2}$) and UNRELATED otherwise; all seven pairs fall unambiguously on the expected side, with the nearest related checkpoint (CodeLlama, $\mathcal{L} = 0.336$) still more than two orders of magnitude above the null.

Pair (base vs.)	Kind	$\mathcal{L}$	W.cos	Verdict
<i>Related (real fine-tunes)</i>				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
<i>Unrelated (cross-model null)</i>				
Mistral-7B-v0.1	scratch	4e−4	0.158	UNR.
OpenLLaMA-7B	scratch [†]	4e−4	0.089	UNR.
Yi-6B	scratch	5e−4	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	4e−4	0.162	UNR.

Table 11: Real-LLM lineage validation on the public LLaMA-2 family (7 real pairs). All seven verdicts match expectation; REL. / UNR. abbreviate RELATED / UNRELATED. [†]OpenLLaMA shares LLaMA-2’s exact architecture ($d=4096$, $L=32$, $d_{\text{ff}}=11008$) but is trained from scratch, so its classification as unrelated cannot come from configuration. Diagonal dominance ($\mathcal{L}$) drives every unrelated checkpoint to the 10^{-4} null while keeping related checkpoints at ≥ 0.34 ; weight cosine (W.cos) also separates but compresses the unrelated only to ≈ 0.13 . CodeLlama, a heavy code-specialized continued-pretraining fine-tune, is the weakest related checkpoint ($\mathcal{L} = 0.336$) yet still $\sim 700\times$ above the null—graceful degradation with fine-tune intensity rather than a sharp threshold. All models share $d = 4096$, $L = 32$; branch products are $d \times d$ regardless of d_{ff} .

Baseline separation. Aggregated over the seven pairs, diagonal dominance separates related from unrelated checkpoints by $\approx 1850\times$ in mean score (0.78 vs. 4.2×10^{-4}), versus $6\times$ for weight cosine (0.80 vs. 0.13); IPGuard, available only on the four-pair logit subset, gives $2\times$ (0.25 vs. 0.13); aligned Frobenius and singular-value distance do not

separate this real-model set (AUROC 0.667 each), as their global spectral/Frobenius statistics are dominated by cross-model scale differences. This reproduces on real 7B weights the margin advantage established on the synthetic and CIFAR benchmarks: several methods clear AUROC on easy splits, but the residual-signature score preserves the largest related-vs-unrelated margin.

The set already spans a real RLHF model, a real instruction fine-tune, and a real continued-pretraining derivative (CodeLlama), tested against the strongest available negative—a same-architecture from-scratch model (OpenLLaMA) whose classification as unrelated cannot be explained by configuration alone. The remaining limitation is tree breadth: all related checkpoints here derive from a single base (LLaMA-2-7B). Extending to additional real checkpoints trees (e.g. a Mistral or Qwen base with its own community fine-tunes) and to a larger negative set that supports a calibrated FPR is the natural next step.

D.5 Harder Regime: Layer Grafts and Linear Merges

To probe the regime predicted by Section 5.4 where a flat global cosine should blur partial provenance, we evaluate all seven baselines on two harder constructions over the depth-16 MLP zoo: (i) **layer grafts**, where a suspect is built by copying $K \in \{0, 4, 8, 12, 16\}$ of the reference’s 16 blocks into a same-task independent donor (and labelling $K \geq 8$ as related); and (ii) **linear merges**, where suspect parameters equal $w \cdot \text{ref} + (1 - w) \cdot \text{donor}$ with $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (related if $w \geq 0.5$). We use 2 trained references, 2 trained donors per reference, and report AUROC for binary related-vs-unrelated detection plus Spearman correlation between each method’s score and the reference fraction (K/L or w); the latter measures whether a method tracks partial overlap monotonically rather than just clearing a threshold. Table 12 summarizes the 40-pair benchmark.

D.6 ResNet Factorization

For Bottleneck blocks, the naive W_3W_1 achieves chance-level accuracy. The triple product $W_3W_2W_1$ recovers:

- ResNet-50/layer3 ($n=5$): 100%, AUC 1.000
- ResNet-101/layer3 ($n=22$): 100%, AUC 0.995
- ResNet-152/layer3 ($n=35$): 91%, AUC 0.964

D.7 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 related checkpoints (noise 1–10%, pruning 20–80%, quantization 16–256 levels). Results: AUROC=1.000, TPR@1%FPR=100%. Related checkpoint scores: $\mathcal{L} \in [0.695, 1.000]$; unrelated baseline: $\mathcal{L}_{\text{max}} = 0.004$.

Method	AUROC		Spearman ρ	
	Graft	Merge	Graft	Merge
Diagonal dominance (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (regr. analogue)	0.052	0.448	-0.83	+0.22

Table 12: Harder-regime benchmark on layer grafts and linear merges (40 pairs total). Related is defined as $K \geq L/2$ for grafts and $w \geq 0.5$ for merges. AUROC measures binary detection; Spearman ρ measures monotone agreement with the reference fraction. Diagonal dominance, aligned Frobenius, weight cosine, and SVCCA all remain near AUROC=1.000 with $\rho \geq 0.96$ on both regimes—confirming Section 5.4’s framing that the contribution is *not* that residual-product dominance beats generic weight cosine on aggregate detection. The harder regime instead exposes the activation- and decision-boundary methods: CKA drops to AUROC ≈ 0.80 with $\rho \approx 0.74$, IPGuard collapses entirely (AUROC 0.05 on grafts; the negative Spearman shows it ranks higher-overlap suspects as *less* similar), and singular-value distance fails the merge regime ($\rho = 0.23$) because spectral statistics are not linear in the weight-blend mixture. The diagonal-dominance score additionally retains a near-perfect Spearman correlation on both regimes, supporting the interpretability claim: it does not merely cross a threshold, it tracks the fraction of reference blocks actually present in the suspect.

D.8 Branching Ancestry Recovery

On a synthetic tree $C_1 \rightarrow \{C_2, C_3\}$, $C_2 \rightarrow C_4$, $C_3 \rightarrow C_5$, the lineage score ranks candidates correctly:

$$\begin{aligned} \mathcal{L}(C_4, C_2) = 0.961 &> \mathcal{L}(C_4, C_1) = 0.910 \\ &> \mathcal{L}(C_4, C_3) = 0.867 > \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

E Attack and Robustness Details

E.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \mathcal{L}_{\cos}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (14)$$

E.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 13: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq 12\%$ utility loss.

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

F Practical Applications

The fingerprint enables applications beyond forensic verification.

F.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $< 50\%$ at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

F.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\varepsilon I + E$) provides a commitment scheme: owners commit to $\text{Hash}(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

F.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E -matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.

G Extended Limitations

This appendix provides detailed discussion of limitations summarized in the main text.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as unrelated despite functional similarity. The method cannot detect training-data overlap or knowledge transfer—if two models were trained on the same data from different initializations, they are unrelated by our definition even though they may exhibit similar behavior.

Symmetric, no direction. $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol establishes *whether* two checkpoints share weight lineage but not *which* is the ancestor. Directional attribution requires either training-time metadata (timestamps, version numbers) or comparison against a third candidate. The branching-tree experiment (Section C) shows that comparing against multiple candidates can recover ancestry order: parent scores higher than grandparent, which scores higher than uncle.

Real-LLM validation scope. The 7-pair LLaMA-2 test demonstrates descriptive separation ($\approx 1850\times$) rather than calibrated false-positive rate. With only four real independents, the null distribution is too small for statistical confidence. The remaining gaps:

- Tree breadth: all related checkpoints derive from LLaMA-2-7B. Extending to Mistral or Qwen bases with their own community fine-tunes would test generalization across foundation-model families.
- Negative set size: a calibrated FPR requires $\sim 50+$ independents. The per-block branch products fit comfortably in memory at 7B scale, so this is engineering rather than methodological.

Adaptive attack coverage. The gradient-based suppression attack (Section 6.2) is the only adaptive attack tested. The attacker has white-box knowledge of the score function, which is a strong threat model, but alternative attack surfaces remain unexplored:

- Git Re-Basin-style re-alignment toward a third model could potentially rotate the E matrices away from their checkpoint-specific directions.
- Extended fine-tuning with an explicit E -decorrelation term (minimizing $\langle E_{\text{suspect}}, E_{\text{reference}} \rangle$) is a stronger, fingerprint-aware version of the tested attack.
- Warm-start distillation (student initialized from teacher weights, then distilled) is by our Section 2 definition RE-LATED, making it a positive control that sharpens the distillation boundary rather than an evasion path.

Cross-architecture distillation. DistilBERT from BERT, or any student with different depth or width than its teacher, is out of scope. The protocol returns INCOMPATIBLE when architectures differ. Extending to cross-architecture comparison would require either (a) a learned alignment between architectures, or (b) a representation-space comparison that abandons the weight-only constraint.

Non-residual architectures. The fingerprint depends on the skip connection creating a functional constraint toward identity. MLP-Mixer, pure attention without residual connections, or any architecture where $x_{\ell+1} = F_{\ell}(x_{\ell})$ rather than $x_{\ell+1} = x_{\ell} + F_{\ell}(x_{\ell})$ will not develop diagonal-dominant branch products. The PlainNet control (Section 5.1) confirms this: architecturally identical to ResNet except for removed skip connections, it achieves only 3% pair accuracy at the chance baseline.